From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

FIITJEE

JEE (Main), 2014

FULL TEST – II

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

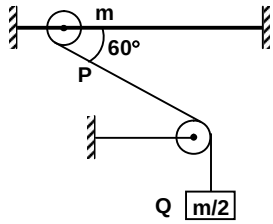
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A swimmer can swim in still water with a speed of $\sqrt{5}$ m/s. While crossing a river his average speed is 3 m/s. If he cross the river in the shortest possible time, what is the speed of flow of water?
(A) 2 m/s (B) 4 m/s
(C) 6 m/s (D) 8 m/s

2. A car starting from rest is accelerated at constant rate until it attains a constant speed v. It is then retarded at a constant rate until it comes to rest. Considering that the car moves with constant speed for half of the time of total journey, the average speed of the car for the journey is
(A) $\frac{v}{4}$ (B) $\frac{3v}{4}$
(C) $\frac{3v}{2}$ (D) Data insufficient

3. A smooth ring P of mass m can slide on a fixed horizontal rod. A string tied to the ring passes over a fixed pulley and carries a block Q of mass (m/2) as shown in the figure. At an instant, the string between the ring and the pulley makes an angle 60° with the rod.
The initial acceleration of the ring is

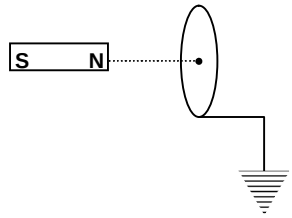


(A) $\frac{2g}{3}$ (B) $\frac{2g}{6}$
(C) $\frac{2g}{9}$ (D) $\frac{g}{3}$

4. A block of mass M is hanging over a smooth and light pulley through a light string. The other end of the string is pulled by a constant force F. If kinetic energy of the block increases by 20 J in 1s. Then
(A) tension in the string is Mg.
(B) tension in the string is F
(C) Work done by the tension on the block is 20 J in 1 sec.
(D) Work done by the force of gravity is 20 J in 1 sec.

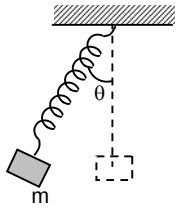
Space for Rough work

5. A circular ring is fixed in a gravity free space and one point of the ring is earthed. Now a magnet is placed along axis of the ring at a distance from its centre such that the nearer pole is north pole as shown in figure. A sharp impulse is applied on the magnet so that it starts to move towards the ring. Then,



(A) Initially magnet experiences an acceleration and then it retards to come to an instantaneous rest.
(B) Magnet starts to oscillate about centre of the ring
(C) Magnet continuous to move along the axis with constant velocity
(D) The magnet retards and comes to rest finally.

6. A block of mass m is suspended by means of an ideal spring of force constant k from ceiling of a car which is moving along a circular path of radius 'r' with acceleration 'a'. The time period of oscillation of the block when it is displaced along the spring, will be



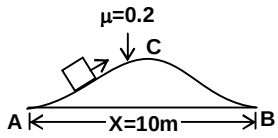
(A) $2\pi\sqrt{\frac{mg+ma}{k}}$ (B) $2\pi\sqrt{\frac{m}{k\sqrt{g^2+a^2}}}$

(C) $2\pi\sqrt{\frac{m}{k}}$ (D) $2\pi\sqrt{\frac{m}{\sqrt{k^2+g^2+a^2}}}$

7. In the hydrogen atom spectrum λ_{3-1} and λ_{2-1} represent wavelengths emitted due to transition from second and first excited states to the ground state respectively. The value of $\frac{\lambda_{3-1}}{\lambda_{2-1}}$ is

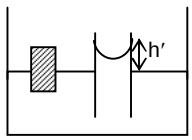
(A) 27 / 32 (B) 32/27
(C) 4/9 (D) 9/4

8. A block of mass 1 kg is pulled along the curve path ACB by a tangential force as shown in figure. The work done by the frictional force when the block moves from A to B is



(A) 5 J (B) 10 J
(C) 20 J (D) none of these

9. In a capillary tube placed inside the liquid of density (ρ)in a container, the rise of liquid is h. When block of density 'σ' is placed on the liquid as shown in figure, liquid in the tube is h′. If $\sigma < \rho$ then



(A) h′ = h (B) h′ < h
(C) h′ > h (D) insufficient data

Space for Rough work

10. A light rod of length L, is hanging from the vertical smooth wall of a vehicle moving with acceleration $\sqrt{3}$ g having a small mass attached at it's one end is free to rotate about an axis passing through the other end. The minimum velocity given to the mass at it's equilibrium position so that it can complete vertical circular motion is

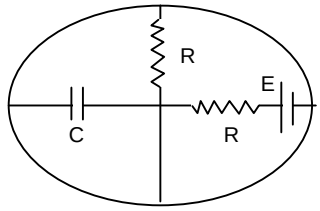
(A) $\sqrt{5gL}$ (B) $\sqrt{4gL}$

(C) $\sqrt{8gL}$ (D) none of these

11. Energy stored in the capacitor in it's steady state is

(A) CV (B) $\frac{CV}{2}$

(C) $\frac{1}{2}QV$ (D) zero



12. The potential difference between two points A and B which are separated by a distance of 1m along the field in a uniform electric field of $10NC^{-1}$ is

(A) zero (B) 100 volt

(C) 10 volt (D) 0.1 volt

13. The power factor of a circuit in which a box having unknown electrical devices connected in series with a resistor of resistance 3Ω is 3/5. The reactance of the box is

(A) $5\ \Omega$ (B) $5/3\ \Omega$

(C) $4\ \Omega$ (D) $4/3\ \Omega$

14. Two points A and B are at distances of 'a' and 'b' respectively from an infinite conducting plate having charge density σ. The work done in moving charge Q from A to B is

(A) $\frac{Q\sigma}{\varepsilon_0}(b-a)$ (B) $\frac{\sigma}{(b-a)}Q$

(C) $\frac{Q\sigma}{(b-a)\varepsilon_0}$ (D) none of these

15. A point charge of 0.1C is placed on the circumference of a non-conducting ring of radius 1m which is rotating with a constant angular acceleration of 1 rad/sec^2. If ring starts it's motion at t = 0 the magnetic field at the centre of the ring at t = 10 sec, is

(A) 10^{-6} T (B) 10^{-7} T

(C) 10^{-8} T (D) 10^{7} T

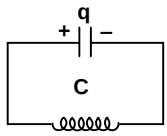
Space for Rough work

16. The wavelength corresponding to maximum spectral radiancy of a black body A is λ_A = 5000 Å. Consider another black body B whose surface area is twice of that of A and total radiant energy emitted by B is 16 times that emitted by A. The wavelength corresponding to maximum spectrum radiancy for B will be
(A) 5000 $(8)^{1/4}$ Å (B) 2500 Å
(C) 10, 000 Å (D) $\frac{5000}{(8)^{1/4}}$ Å

17. During an adiabatic process, the density of a gas is found to be proportional to cube of temperature. The degree of freedom of gas molecule is
(A) 6 (B) 5
(C) 4 (D) 3

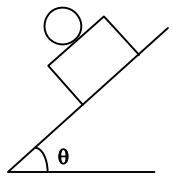
18. Two fixed charges $-2Q$ and Q are located at the point of co-ordinates $(-3a, 0)$ and $(3a, 0)$ respectively in $(x - y)$ plane. Then all the points in x-y plane where potential is zero lies on a
(A) straight line parallel to x-axis (B) straight line parallel to y- axis
(C) a circle of radius 4a (D) circle of radius 2a

19. In an L-C circuit shown in the figure, C = 1F, L = 4H. At time t = 0, charge in the capacitor is 4C and it is decreasing at a rate of $\sqrt{5}$ C/s. Choose the correct statements.



(A) maximum charge in the capacitor can be 6C
(B) maximum charge in the capacitor can be 8C
(C) charge in the capacitor will be maximum after time 2 $\sin^{-1}(2/3)$ sec
(D) None of these

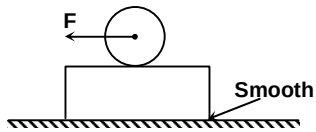
20. A plank of mass M is placed over smooth inclined plane and a sphere is also placed over the plank. Friction is sufficient between the sphere and the plank to prevent slipping. If the plank and the sphere are released from rest the frictional force on the sphere is



(A) up the plane (B) down the plane
(C) zero (D) horizontal

Space for Rough work

21. A disc of mass m and radius R is placed over a plank of same mass m. There is sufficient friction between the discs and the plank to prevent slipping. A force F is applied at the centre of the disc. Choose the correct statements.



(A) Acceleration of the plank is $\frac{F}{4m}$

(B) Acceleration of the plank is $\frac{F}{2m}$

(C) Force of friction between disc and plank is $\frac{F}{6}$

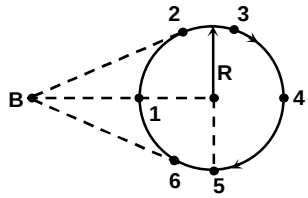
(D) Force of friction between disc and plank is $\frac{F}{2}$

22. A certain amount of ideal monoatomic gas undergoes, process given by $UV^{1/2} = C$ where U is the internal energy of the gas. The molar specific heat of the gas for the process will be

(A) R/2 (B) 3R

(C) 5R/2 (D) –R/2

23. An isotropic sound source A is moving in a circle of radius R with a small speed v. An observer B is hearing this sound (See figure). The intensity of the sound heard by B will be maximum when the source is at point.



(A) 1 (B) 2

(C) 6 (D) none of these

24. A stable species C can be obtained through unstable nuclides A and B. The half life period for conversion A $\rightarrow$ C is T and that for conversion B $\rightarrow$ C is 2T. Initially a sample contains N nuclides of A, N nuclides of B and no nuclides of C. After how much time will the nuclides of species C be equal to (27/16) N

(A) 2T (B) 3T

(C) 4T (D) 5T

25. A certain thermodynamic cycle comprises of two isothermal and two adiabatic processes. The highest temperature obtained in the entire cycle is 600 K while the lowest temperature is 300 K. When the cycle represented on a P-V curve the sense of the cycle is counter clockwise. The efficiency of the cycle is

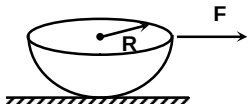
(A) 50% (B) 75 %

(C) 25 % (D) None of these.

Space for Rough work

26. A man wearing spectacles with diverging lenses is viewing a distance object. The image formed on his retina will be
(A) real and erect. (B) real and inverted
(C) virtual and erect (D) virtual and inverted.

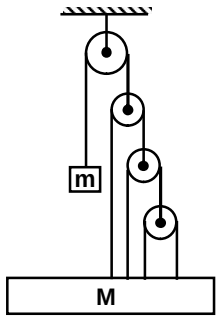
27. A thin uniform hemispherical bowl of mass m and radius R is lying on a smooth horizontal surface. A horizontal force F is now applied perpendicular to the rim of the bowl (see figure). The instantaneous angular acceleration of the bowl will be



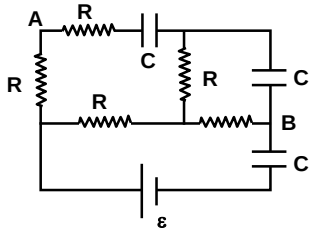
(A) $\frac{20}{3}\frac{F}{MR}$ (B) $\frac{10}{3}\frac{F}{MR}$
(C) $\frac{40}{3}\frac{F}{MR}$ (D) $\frac{5}{3}\frac{F}{MR}$

28. For a stone thrown from a tower of unknown height, the maximum range for a projection speed of 10 m/s is obtained for a projection angle of 30°. The corresponding distance between the foot of the tower and the point of landing of the stone is
(A) 10 m (B) 20 m
(C) $\left(20/\sqrt{3}\right)$ m (D) $\left(10/\sqrt{3}\right)$ m

29. Consider the system shown in the figure. the system is so arranged that both m and M are in pure translation. If at a certain instant M is moving down with a speed of 1 m/s then the speed of m will be
(A) 7 m/s (B) 8 m/s
(C) 1 m/s (D)15 m/s



30. Consider the circuit in the adjacent figure. What will be potential difference between A and B in the steady state
(A) ε (B) $\varepsilon/2$
(C) $\varepsilon/3$ (D) zero



Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. For the equilibrium, $Ag^+ + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+$ $K_1 = 1.8 \times 10^7$ and $Ag^+ + Cl^- \rightleftharpoons AgCl$; $K_2 = 5.6 \times 10^9$ hence, for the equilibrium : $AgCl + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+ + Cl^-$ equilibrium constant is :

   (A) 0.32×10^{-2} (B) 0.31×10^{-21}
   (C) 1.01×10^{17} (D) 1.01×10^{-17}

2. 2-Pentyne, on reduction with lithium in liq. NH_3 yields

   (A) pentane (B) cis-2-pentene
   (C) trans-2-pentene (D) 1-pentene

3. $A(C_7H_7Cl) \xrightarrow[\text{(ii) Sodalime}/\Delta]{\text{(i) } KMnO_4}$ Chlorobenzene. A forms only one nitro derivative. Hence A is,

   (A) Benzyl chloride (B) p-Chlorotoluene
   (C) o-Chlorotoluene (D) m-Chlorotoluene

4. Claisen rearrangement of allyl phenyl ether gives a mixture of
   (A) o-, m-and p-allyl phenols. (B) o-and m-allyl phenols.
   (C) m-and p-allyl phenols. (D) o-and p-allyl phenols.

5. If in diamond, there is a unit cell of carbon atoms as fcc and if carbon atom is sp^3 hybridized, what fractions of void are occupied by carbon atom.
   (A) 25% tetrahedral (B) 50% tetrahedral
   (C) 25% octahedral (D) 50% octahedral

Rough Work

6. Out of boiling point (I), entropy (II), pH (III) and standard e.m.f. of a cell (IV), intensive properties are -
(A) I, II (B) I, II, III
(C) I, III,IV (D) all of these

7. Which metal liberates H_2 with very dilute nitric acid:
(A) Zn (B) Cu
(C) Mn (D) Hg

8. Which of the following statements are not correct ?
(A) The number of α and β-particles emitted in neptunium series $({}^{237}_{93}Np \rightarrow {}^{209}_{83}Bi)$ are 7 and 4, respectively.
(B) The number of α and β-particles emitted in uranium series $({}^{238}_{92}U \rightarrow {}^{206}_{83}Pb)$ are 8 and 7, respectively.
(C) The number of α and β-particles emitted in actinium series $({}^{235}_{92}\text{U} \rightarrow {}^{207}_{83}\text{Bi})$ are 4 and 7, respectively.
(D) The half-life of a radioactive decay is given as $t_{1/2} = 0.693/\lambda$, where λ is decay constant

9. Reactivity of borazole is greater than that of benzene because:
(A) borazole is non-polar compound (B) borazole is polar compound
(C) borazole has electron in it (D) of localized electrons in it

10. When acetone and chloroform are mixed, hydrogen bonding takes place between them. Such a liquid pair will cause.
(A) No deviations from Raoult's law (B) Negative deviations from Raoult's law
(C) Positive deviations from Raoul'ts law (D) None

Rough Work

11. On heating ammonium dichromate, the gas evolved is
(A) Oxygen (B) Ammonia
(C) Nitrous oxide (D) Nitrogen.

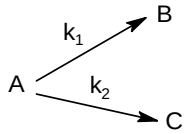
12. When ammonia is added to a cupric salt solution, the deep blue colour, so observed is due to the formation of
(A) $[Cu(OH)_4]^{2-}$ (B) $[Cu(NH_3)_4]^{2+}$
(C) $[Cu(OH)_2(NH_3)_2]$ (D) $[Cu(H_2O)_4]^{2+}$

13. Arrange the following in order of their decreasing thermal conductivity
(A) Al, Ag, Cu (B) Cu, Ag, Al
(C) Ag, Cu, Al (D) Al, Cu, Ag.

14. In a measurement of quantum efficiency of photo-synthesis in green plants, it was found that 10 quanta of red light of wavelength 6850 $\overset{o}{A}$ were needed to release one molecule of O_2. The average energy storage in this process is 112 kcal/mol O_2 evolved. What is the energy conversion efficiency in this experiment? Given 1 cal = 4.18 J, $N_A = 6 \times 10^{23}$, $h = 6.64 \times 10^{-34}$ JS.
(A) 23.5 (B) 26.9
(C) 66.34 (D) 73.1

15. The rate of effusion of two gases 'a' and 'b' under identical condition of temperature and pressure are in the ratio of 2 : 1. What is the ratio of rms velocity of their molecules if T_a and T_b are in the ratio of 2 : 1?
(A) 2 : 1 (B) $\sqrt{2}:1$
(C) $2\sqrt{2}:1$ (D) $1:\sqrt{2}$

16. For first order parallel reaction k_1 and k_2 are 4 and 2 min^{-1} respectively at 300 K. If the activation energies for the formation of B and C are respectively 30,000 and 33,314 J/mol respectively. The Temperature at which B and C will be obtained in equimolar ratio is:



(A) 757.48 K (B) 329.77 K
(C) 600 K (D) None of these

Rough Work

17. One mole of SO_3 was placed in a two litre vessel at a certain temperature. The following equilibrium was established in the vessel.

$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

The equilibrium mixture reacted with 0.2 mole $KMnO_4$ in acidic medium. Hence K_c is:

(A) 0.50 (B) 0.25
(C) 0.125 (D) none of these

18. A complex of iron and cyanide ions is 100% ionized at 1 molal. If its elevation in boiling point is 2.08^o, then the complex is: (Given $K_b = 0.52^oC\ mol^{-1}\ kg$)

(A) $Fe_4[Fe(CN)_6]_3$ (B) $K_3[Fe(CN)_6]$
(C) $K_4[Fe(CN)_6]$ (D) $Fe_3[Fe(CN)_6]_2$

19. Find the standard cell potential involving the cell reaction:

$In^{2+} + Cu^{+2} \longrightarrow In^{+3} + Cu^{+}$, at 298 K

Given: $E^0_{Cu^{+2}/Cu^+} = X_1V$, $E^0_{In^{+3}/In^+} = X_2V$ $E^o_{In^{+2}/In^+} = X_3V$

(A) $X_1 + X_3 - X_2$ (B) $(X_1 + X_3 - 2X_2)/3$
(C) $X_1 + X_3 - 2X_2$ (D) $X_1 + X_3 + 2X_2$

20. $SnCl_4$ is a covalent liquid because

(A) e^- clouds of the Cl^- ions are weakly polarized to envelop the cation
(B) electron clouds of the Cl^- ions are strongly polarized to envelop the cation
(C) its molecules are attracted to one another by strong vander Waal's forces
(D) Sn shows inert-pair effect

21. Boron nitride obtained by heating borazole is:

(A) white solid with a diamond like structure
(B) slippery white solid with layered structure similar to that of graphite
(C) covalent liquid and is structurally similar to carbon monoxide
(D) low melting solid with rock-salt like structure molecules

22. If a metal has low oxygen affinity then purification of metal may be carried out by

(A) Liquation (B) Distillation
(C) Zone refining (D) Cupellation

Rough Work

23. An original salt solution in acidic medium did not give any precipitate on passing H_2S gas. Such a solution was boiled, reboiled after dilution 3 times. To such a solutions two drops of conc. HNO_3 were added, then heated and water was added. To this resulting solution, NH_4Cl was first added followed by excess of NH_4OH. Finally a green ppt. was obtained. Hence the cation may be:
(A) Al^{+3} (B) Fe^{+2}
(C) Fe^{+3} (D) Cr^{+3}

24. Which of the following statement is wrong?
(A) d orbital taking part in dsp^2 is $d_{x^2-y^2}$
(B) d orbital taking part in sp^3d is d_{yz}
(C) d orbitals taking part in sp^3d^2 are $d_{x^2-y^2}$ and d_{z^2}
(D) d orbitals taking part in sp^3d^3 are d_{xy}, d_{z^2} and $d_{x^2-y^2}$

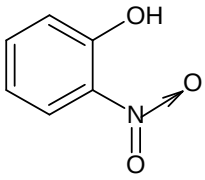
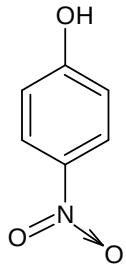
25. $H_3CCOOH \xrightarrow{Br_2/P} (Y) \xrightarrow[(ii)\ H_3O^+]{(i)\ KCN} (X)$
Here (X) is:
(A) Glycollic acid (B) α-hydroxypropionic acid
(C) succinic acid (D) malonic acid

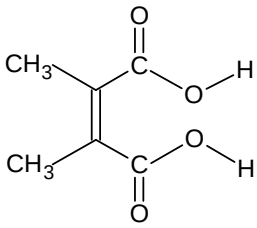
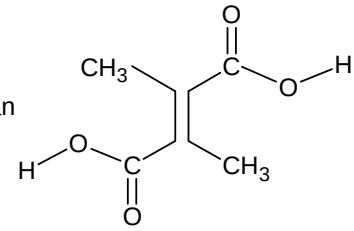
26. In order to distinguish between $C_2H_5NH_2$ and $C_6H_5NH_2$, which of the following reagents is useful
(A) Heinsberg reagent (B) p-napthol
(C) Benzene diazonium chloride (D) None of these

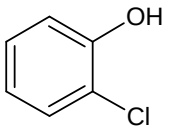
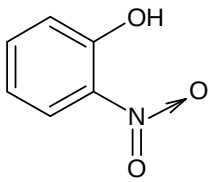
27. The weakest interparticle forces are present in
(A) thermosetting polymers (B) thermoplastic polymers
(C) fibres (D) elastomers

Rough Work

28. Which of the following statement is false ?

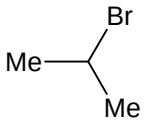
(A)  has lower boiling point than 

(B)  is less stable than 

(C)  has greater pKa than 

(D) H_3BO_3 in C_2H_5OH exists in the cage form due to intermolecular hydrogen bonding.

29. Which among the following compounds will not give effervescence with sodium bicarbonate?
(A) C_6H_5COOH (B) $C_6H_5SO_3H$
(C) C_6H_5OH (D) picric acid

30. Which of the following is most reactive for S_N2 reaction?
(A)  (B) 
(C)  (D) 

Rough Work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice questions** numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Sum of all the roots of the equation $x^2 - 2x + |x - 1| - 5 = 0$ is
   (A) 0 (B) 2
   (C) 1 (D) 5

2. If z_1, z_2, z_3, z_4 are points on the circle $|z| = 1$ such that $z_1 + z_2 + z_3 + z_4 = 0$ and the value of the expression $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_4|^2 + |z_4 - z_1|^2$ is least then,
   (A) $z_1 + z_3 = 0$ and $z_2 + z_4 = 0$ (B) $z_1 + z_2 = z_3 + z_4$
   (C) z_1, z_2, z_3, z_4 must be real (D) $\frac{z_1}{z_3} = \frac{z_2}{z_4}$

3. If $T_r = \frac{r}{r^4 + 4}$ and $S_n = \sum_{r=1}^{n} t_r$, then the value of $37S_5 - \frac{7}{26}$ is equal to
   (A) 13 (B) $\frac{25}{26}$
   (C) $\frac{1}{2}$ (D) $\frac{7}{26}$

4. Number of distinct terms in the expansion of $(x + y + z + w)^{50}$ is equal to
   (A) $^{53}C_3$ (B) 51
   (C) 51^2 (D) $^{51}C_3$

5. If $y = f(x)$ is symmetric about the lines $3x + 4y + 1 = 0$ and $4x - 3y - 7 = 0$, then it must be symmetric about
   (A) (1, 1) (B) (1, –1)
   (C) (0, 0) (D) (1, 0)

Space for rough work

6. If $\sqrt{1-x^6}+\sqrt{1-y^6}=a\left(x^3-y^3\right)$ and $\frac{dy}{dx}=f(x, y)\sqrt{\frac{1-y^6}{1-x^6}}$ then,

(A) $f(x, y)=\frac{y}{x}$ (B) $f(x, y)=\frac{x^2}{y^2}$

(C) $f(x, y)=\frac{2y^2}{x^2}$ (D) $f(x, y)=\frac{y^2}{x^2}$

7. If with standard notations t_1, t_2, t_3, t_4 are four co–normal points on the hyperbola $xy = c^2$ then the orthocentre of the triangle formed by joining the points t_1, t_2, t_3 is given by

(A) (0, 0) (B) $\left(ct_4, \frac{c}{t_4}\right)$

(C) $\left(c\left(t_1+t_2+t_3\right), \frac{c}{t_1+t_2+t_3}\right)$ (D) $\left(\frac{c}{t_4}, ct_4\right)$

8. If S be the focus of a parabola and PQ be the focal chord, such that SP = 3 and SQ = 6, then the length of latus rectum of the parabola is

(A) 4 (B) 2

(C) 8 (D) 16

9. If a circle having centre at (α, β) cut the circles $x^2 + y^2 – 2x – 2y – 7 = 0$ and $x^2 + y^2 + 4x – 6y – 3 = 0$ orthogonally, then $\left|\frac{3}{4}\alpha-\frac{\beta}{2}\right|$ is equal to

(A) 1 (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) 0

10. Two circles with radii 'r_1' and 'r_2', $r_1 > r_2 \geq 2$, touch each other externally. If 'θ' be the angle between the direct common tangents, then

(A) $\theta=\sin^{-1}\left(\frac{r_1+r_2}{r_1-r_2}\right)$ (B) $\theta=2\sin^{-1}\left(\frac{r_1-r_2}{r_1+r_2}\right)$

(C) $\theta=\sin^{-1}\left(\frac{r_1-r_2}{r_1+r_2}\right)$ (D) none of these

Space for rough work

11. A unit vector in xy – plane which makes an angle of 45° with the vector $\vec{i}+\vec{j}$ and an angle of 60° with the vector $3\vec{i}-4\vec{j}$ is

(A) $\vec{i}$ (B) $\frac{\vec{i}+\vec{j}}{\sqrt{2}}$

(C) $\frac{\vec{i}-\vec{j}}{\sqrt{2}}$ (D) none of these

12. Let $(1+x)^n = 1 + nx + \frac{n(n-1)x^2}{2} + \ldots . x$, $n \in R$. Then the sum of the series $3+\frac{8}{3}+\frac{80}{3^3}+\frac{240}{3^4}+\ldots.$ is

(A) 9 (B) 27
(C) 12 (D) 101

13. There are three points (a, x), (b, y) and (c, z) such that the straight lines joining any two of them are not equally inclined to the coordinate axes where a, b, c, x, y, $z \in R$.

If $\begin{vmatrix} x+a & y+b & z+c \\ y+b & z+c & x+a \\ z+c & x+a & y+b \end{vmatrix} = 0$ and a + c = –b, then x, $-\frac{y}{2}$, z are in

(A) A.P. (B) G.P.
(C) H.P. (D) none of these

14. If at x = 1, y = 2x is tangent to the parabola $y = ax^2 + bx + c$, then respective values of a, b, c are

(A) $\frac{1}{2}, 1, \frac{1}{2}$ (B) 1, $\frac{1}{2}$, $\frac{1}{2}$

(C) $\frac{1}{2}, \frac{1}{2}, 1$ (D) none of these

15. If $f'\left(x^2-4x+3\right) > 0$, $\forall x \in (2, 3)$; then f(sin x) is increasing on

(A) $\bigcup_{n\in I}\left(2n\pi, (4n+1)\frac{\pi}{2}\right)$ (B) $\bigcup_{n\in I}\left((4n-1)\frac{\pi}{2}, 2n\pi\right)$

(C) R (D) none of these

Space for rough work

16. Coordinates of the point on the straight line x + y = 4, which is nearest to the parabola $y^2 = 4(x - 10)$ is

(A) $\left(\frac{17}{2}, -\frac{9}{2}\right)$ (B) (2, 2)

(C) $\left(\frac{3}{2}, \frac{5}{2}\right)$ (D) none of these

17. If $\bar{a}$, $\bar{b}$, $\bar{c}$ be three vectors of magnitude $\sqrt{3}$, 1, 2 such that $\bar{a} \times \left(\bar{a} \times \bar{c}\right) + 3\bar{b} = 0$, if θ is the angle between $\bar{a}$ and $\bar{c}$, then $\cos^2 \theta$ is equal to

(A) $\frac{3}{4}$ (B) $\frac{1}{2}$

(C) $\frac{1}{4}$ (D) none of these

18. If $\Delta(x) = \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \ln(1+x) & \cos x & \sin x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix}$ = A + Bx + Cx^2 + ……., then B is equal to

(A) 0 (B) 1

(C) 2 (D) none of these

19. If the function f : [2, ∞) $\rightarrow$ [1, ∞) is defined by $f(x) = 3^{x(x-2)}$, then $f^{-1}(x)$ is

(A) $1 + \sqrt{1+\log_3 x}$ (B) $1 - \sqrt{1+\log_3 x}$

(C) $1 + \sqrt{1-\log_3 x}$ (D) does not exist

20. The number of real solutions of the equations $\tan^{-1}\sqrt{x^2 - 3x + 2} + \cos^{-1}\sqrt{4x - x^2 - 3} = \pi$

(A) one (B) two

(C) zero (D) infinite

21. Let $x = (5\sqrt{2} + 7)^{19}$, then x{x} ({.} denotes the fractional part of x) is equal to

(A) 2^{19} (B) 3^{19}

(C) 0 (D) 1

Space for rough work

22. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar, is

(A) $\sqrt{3} : 1$ (B) $1 : 3$

(C) $1 : \sqrt{3}$ (D) $\sqrt{3} : 2$

23. $\frac{1}{x} = \frac{2}{3!}e + \frac{4}{5!}e + \frac{6}{7!}e + \ldots \infty$, then find $\int_0^x f(y)\log_y x \, dy$, $y > 1$

(A) $\frac{[f(e)]^2}{2}$ (B) $\frac{\left[f\left(\frac{1}{e}\right)\right]^2}{2}$

(C) $\frac{\left[f(e^2)\right]^2}{2}$ (D) none of these

24. If lines $x = y = z$, $x = \frac{y}{2} = \frac{z}{3}$ and third line passing through (1, 1, 1) form a triangle of area $\sqrt{6}$ units then point of intersection of third line with second line will be

(A) (1, 2, 3) (B) (2, 4, 6)

(C) $\left(\frac{4}{3}, \frac{8}{3}, \frac{12}{3}\right)$ (D) none of these

25. If $A(\bar{a})$, $B(\bar{b})$ and $C(\bar{c})$ be vertices of a triangle whose circumcentre is the origin, then orthocentre is given by

(A) $\frac{\bar{a}+\bar{b}+\bar{c}}{3}$ (B) $\frac{\bar{a}+\bar{b}+\bar{c}}{2}$

(C) $\bar{a}+\bar{b}+\bar{c}$ (D) none of these

26. Let $f(x) = \max\{\tan x, \cot x\}$. Then number of roots of the equation $f(x) = \frac{1}{\sqrt{3}}$ in $(0, 2\pi)$ is

(A) 2 (B) 4

(C) 0 (D) infinite

Space for rough work

27. If S denote the sum to infinity and S_n the sum of n terms of the series $1+\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\ldots$ such that $S - S_n < \frac{1}{300}$, then the least value of n is

(A) 4 (B) 5
(C) 6 (D) 7

28. If $f(x) = \int_0^4 e^{|t-x|}dt$ $(0 \le x \le 4)$, the maximum value of f(x) is

(A) $e^4 - 1$ (B) $2(e^2 - 1)$
(C) $e^2 - 1$ (D) none of these

29. Let $x, y \in [0, 10]$, then number of solutions (x, y) of the inequation $|3^{\sec^2 x-1}\sqrt{9y^2-6y+2} \le 1$ is

(A) 4 (B) 2
(C) 1 (D) infinite

30. A triangle is inscribed in a circle. The vertices of the triangle divide the circle in to three arcs of length 3, 4 and 5 units, then area of the triangle is equal to,

(A) $\frac{9\sqrt{3}\left(1+\sqrt{3}\right)}{\pi^2}$ (B) $\frac{9\sqrt{3}\left(\sqrt{3}-1\right)}{\pi^2}$

(C) $\frac{9\sqrt{3}\left(1+\sqrt{3}\right)}{2\pi^2}$ (D) $\frac{9\sqrt{3}\left(\sqrt{3}-1\right)}{2\pi^2}$

Space for rough work